Barton doubles L2 cache, slows clock speed

AMD has announced the launch of a new core for their Athlon processors named Barton. It features a total of 640 KB of on-die cache, up from 384 KB on the older Thoroughbred core. The new core features the same 333 MHz front-side bus. While the size of the L1 cache on the processor remains unchanged, the L2 cache has been upped from the earlier 256 KB to a good 512 KB.

The 2800+ with the Barton core runs with a slower clock speed than the 2800+ with the Thoroughbred core, thus allowing AMD to keep the same processor rating despite the increased cache. AMD's rating system is officially based on the performance of current processors relative to AMD's original Athlon core, but the numbers mirror the clock speeds of competing processors from Intel.

The new Barton 2800+ runs at 2.083 GHz, compared with the older 2800+ clock speed of 2.25 GHz. The Barton 3000+ runs at 2.167 GHz. It is made by a 0.13-micron process. The increased cache memory increases the die size from 84 mm^2 to about 101 mm^2.

Rambus develops technology to make chip connections faster

Chipmaker Rambus has readied a technology called Redwood that lets semiconductors exchange data at very fast rates. It performs the same functions as other standards, such as AMD's HyperTransport. It transfers data at 6.4 GHz. The jump in speed comes from an innovative circuit board design. Existing parallel buses can shift data rapidly, and they generally require that the data is synchronised while it travels from one chip to another.

This leads to a design requirement which keeps all the interconnects between various chips equal, depending on how the chips sit on the circuit board. Redwood gets around this drawback by incorporating a technology called FlexPhase that allows data to travel asynchronously. It will be compatible with chips rigged for various other input-output standards. The company will charge fees for the engineering design services, from the company implementing Redwood.

Broadband through your electric switches

For people residing in the far suburbs of the UK, broadband services from Fast Net, a UK-based company, could soon be available via electricity lines. Trials of this technology have already been run in the small Scottish towns of Campbeltown and Crief. They were a huge

ILLUSTRATIONS: Farzana Cooper

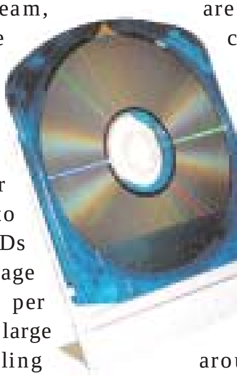
boost for the company with speeds up to 2 MB per second—faster than most broadband services on offer in the UK.

The price for the service could be very affordable. The speed and price of the commercial service has yet to be decided but the firm says it will be competitive.

Blu-ray Disc technology gets the go-ahead

The DVD forum just got better and stronger. Hitachi, LG Electronics, Matsushita Electric Industrial, Pioneer, Royal Philips Electronics, Samsung Electronics, Sharp, Sony and Thomson have announced the arrival of a new format—Blu-ray Disc technology, a next-generation recordable DVD format using blue-violet lasers. Here, data is read off a blue-violet laser beam as opposed to red laser beams used in today's optical drives.

The blue-violet laser beam has a shorter wavelength than a red laser beam, enabling data to be packed with a higher density. This new technology enables storage of 27 GB of data per layer as opposed to conventional DVDs which enable storage of 4.7 GB of data per side. It is seen as a large step towards enabling the recording of high-definition television broadcasts on a DVD platter.



The licensing agreements are 10-year renewable contracts. They will include the right to use the Blu-ray format and logo along with the content protection specifications. Licenses for the format and logo will range from around Rs 10 lakh to Rs 30 lakh depending on the products manufacturers want to develop. The protection specifications range in price annually from Rs 2 lakh to Rs 6 lakh.

Companies are already manufacturing products based on the new technology. Philips demonstrated a prototype Blu-ray disc drive. It uses a 3 cm disc and stores up to 1 GB of data. Typical CDs, measuring 12 cm in diameter, can hold up to 650 MB of data. The prototype drive demonstrated can be used in portable devices such as digital cameras, handhelds and cellphones.

■ US Navy warships to use 802.11b wireless systems ■ NEC demonstrates 50 GFLOPs facial-recognition chip ■ Sony launches all-format DVD-burning Vaio laptops